

MINI-PROJECT 2 SHORT TECHNICAL REPORT

Course: Cross-Platform Mobile App Development (VKU)
Topic: Week 6 - React Native Part 2: Navigation, State, Query, Animations
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Submission Date: 30/09/2026
Live Demo: <https://vku-room-booking.pages.dev/>
Repository: <https://github.com/minhduy6868/mob-lab-room-booking>

1. PROJECT OVERVIEW

VKU Room Booking is a campus room reservation app for labs, seminar rooms, library spaces, makerspaces and smart classrooms. Week 6 upgrades the Week 5 app with typed navigation, separated client/server state, realtime cache sync, native-feeling alerts, Reanimated motion and native gestures.

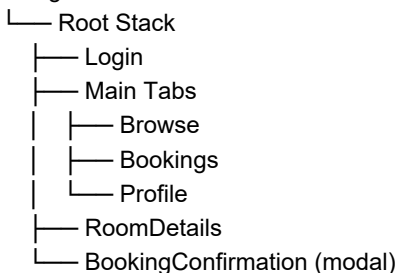
Demo account: abc@vku.udn.vn / vku@2026 (Nguyễn Minh Duy - 23IT038).

2. WEEK 6 IMPLEMENTATION CHECKLIST

Requirement	Status	Implementation
Root Stack + Bottom Tabs	Complete	RootNavigator owns Login, Main tabs, RoomDetails and BookingConfirmation modal.
Type-safe route params	Complete	RootStackParamList defines RoomDetails { roomId, roomName } and BookingConfirmation { bookingId }.
Zustand persist	Complete	Auth session persists; booking store persists selected rooms/bookings/filter cache while KV stays authoritative.
TanStack Query realtime sync	Complete	Live bookings upsert into Zustand and Query cache after POST/PUT; focus/reconnect and pull refresh reconcile KV.
In-app alert UI	Complete	AppAlertProvider replaces browser/native alerts with one polished notice/confirm modal.
Reanimated animations	Complete	Room cards use FadeInDown, FadeOutUp and Layout.springify for polished list transitions.
Gesture Handler	Complete	BookingCard supports swipe-left cancel with Gesture.Pan, shared values and runOnJS.
Code quality	Complete	Strict TypeScript, typed navigation, memoized RoomCard, FlatList windowing and Safe Area setup.

3. NAVIGATION ARCHITECTURE

NavigationContainer



RoomDetails is pushed above the tab navigator so the tab bar is hidden. BookingConfirmation is presented as a modal digital pass after POST /api/bookings succeeds.

4. STATE AND DATA FLOW

Zustand handles client state: auth session, selected date/filter, selected modal room, selected QR pass and cached bookings. TanStack Query handles server state: room catalog and live booking records. Cloudflare Pages Functions with KV remain the source of truth for conflict validation, cancellation and check-in.

The app no longer polls KV on a timer. Successful POST/PUT responses return the changed booking, then the client upserts that record into Zustand and the TanStack Query cache immediately.

Browse card -> RoomDetails -> Booking sheet -> POST /api/bookings

- > server validates conflict rules

- > store/query cache upsert the returned booking record

- > BookingConfirmation modal opens with bookingId

5. MOTION, GESTURE AND DEMO SCRIPT

Room cards animate into the list with Reanimated. Booking cards expose a red cancel action under the card; a left swipe past the threshold triggers the same confirm-safe cancellation path as the visible cancel button.

Demo: login -> tap a room card -> RoomDetails -> book an available slot -> BookingConfirmation pass -> open My Bookings and see the new booking without interval reload -> pull refresh only when reconciling cloud state -> swipe left to cancel -> try an occupied/overlapping slot and verify the 409 message.

6. NOTES

NPM reported existing dependency audit findings after installing Reanimated. No automatic npm audit fix was applied because it may introduce breaking dependency upgrades outside the Week 6 assignment scope.